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Enhancing Subseasonal Temperature Forecasts

via Machine Learning and Nonlinear Dimension Reduction

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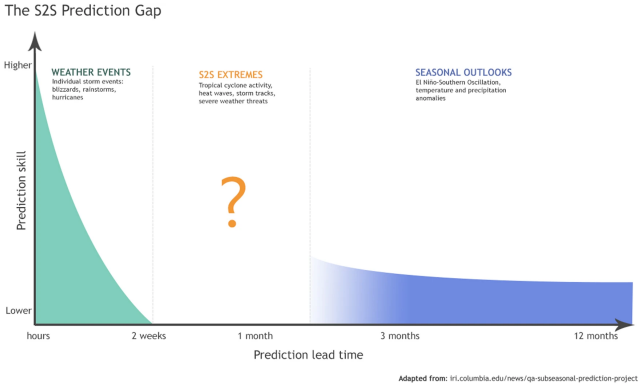
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Motivation

Extreme heat and cold affect energy systems, agriculture, and public health. [1, 2] However, forecast skill from traditional physics-based weather models deteriorates beyond roughly 10–15 days because atmospheric dynamics are chaotic. [5]

Motivation



Source: Mariotti et al. [4].

Goal

Improve subseasonal forecasts of regional temperatures 1–4 weeks ahead.

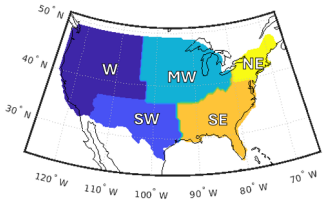
Data and Forecasting Setup

We model residual temperature after removing seasonal and long-term structure:

$$y_{r,t} = \mu_r(t) + \varepsilon_{r,t}$$

- Target Regions: $r \in \{$ W, SW, MW, SE, NE$\}$
- Forecast Horizons: $h = 1, \ldots, 4$ weeks
- $\mathcal{X}_t = \left\{ x_t^{(1)}, \ldots, x_t^{(M)} \right\}$ contains atmospheric fields:
  - Geopotential Height (at 500, 850 hPa)
  - Air Temperature (at 500, 850 hPa)
  - Surface Temperature
  - Sea-Level Pressure
- Input fields are compressed before forecasting

$$x_t^{(m)} \in \mathbb{R}^{p_m}, \qquad z_t \in \mathbb{R}^{k_m}, \qquad k_m \ll p_m.$$



Why Dimension Reduction?

The atmospheric predictor state $x_t^{(m)} \in \mathbb{R}^{p_m}$, is high-dimensional $p_m \gg n$
So we first map atmospheric fields into a lower-dimensional representation:

$$x_t^{(m)} \longmapsto z_t^{(m)}, \qquad z_t^{(m)} \in \mathbb{R}^{k_m}, \qquad k_m \ll p_m.$$

The forecast model uses the latent variables: $\mathbf{z}_t = \left(\left(z_t^{(1)} \right)^\top, \ldots, \left(z_t^{(M)} \right)^\top \right)^\top$

MERRA-2 fields $(\mathcal{X}_t) \rightarrow$ latent features $(\mathbf{z}_t) \rightarrow$ ML Forecast Models $(f) \rightarrow \widehat{\varepsilon}_{r,t+h}$

$$\widehat{\varepsilon}_{r,t+h} = f_{r,h}(\mathbf{z}_t)$$

- Initial work used PCA-derived features.
- This talk compares PCA features with autoencoder-derived features.

Autoencoders as Reconstruction-Based Features

PCA uses a linear encoding of the data: $z_t^{(m)} = V_k^\top \left(x_t^{(m)} - \bar{x}^{(m)} \right)$

An autoencoder replaces this with nonlinear learned maps:

$$z_t^{(m)} = g_{1,\theta_m}\left(x_t^{(m)}\right), \qquad \hat{x}_t^{(m)} = g_{2,\phi_m}\left(z_t^{(m)}\right)$$

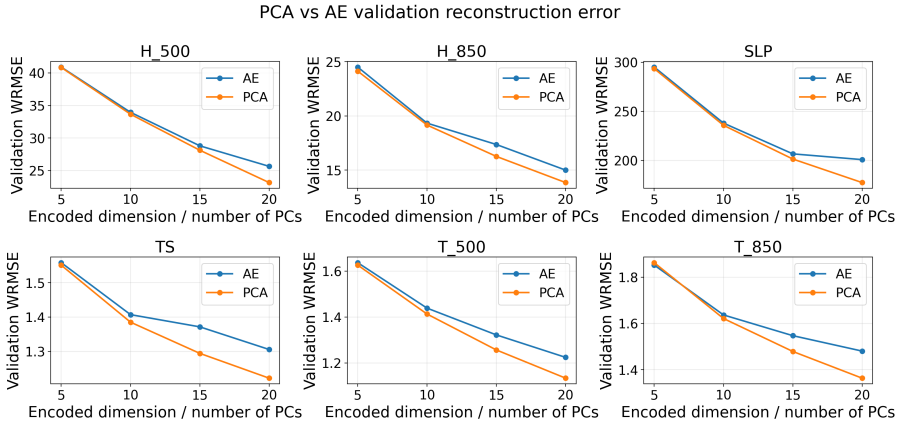
$$(\hat{\theta}_m, \hat{\phi}_m) = \arg\min_{\theta,\phi} \sum_t \left\| D_m \left[x_t^{(m)} - \hat{x}_t^{(m)} \right] \right\|^2 .$$

Where D_m contains location-specific weights: $D_m = \mathrm{diag}\left(\sqrt{w_1^{(m)}}, \ldots, \sqrt{w_{p_m}^{(m)}} \right)$,

Note

In both cases, the latent representation is optimized for reconstructing the input fields $x_t^{(m)}$, not directly for predicting $y_{r,t+h}$.

Reconstruction Comparison



Reconstruction quality evaluates representation fidelity. It does not necessarily imply forecast skill.

Forecast Models

After dimension reduction, we considered two forecast models.

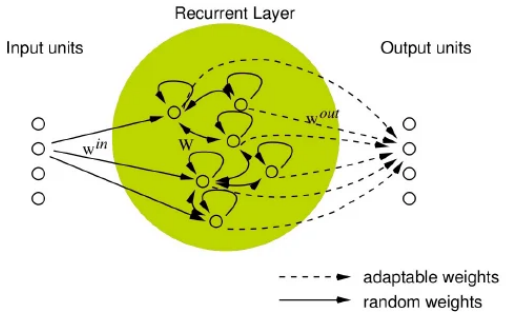
Random Forests

- Ensemble of regression trees.
- Handles nonlinear interactions among latent predictors.
- Strong baseline in prior subseasonal forecasting work.

$$\widehat{\varepsilon}_{r,t+h} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{z}_t).$$

Ensemble Echo State Networks [3, 6, 7, 8]

- Ensemble of Echo State Networks.
- Type of "Reservoir Computing:"



Source: Obst & Boedecke [8].

Echo State Network Details

$$\widehat{\varepsilon}_{r,t+h} = \mathbf{B}\boldsymbol{s}_t, \qquad \boldsymbol{s}_t = g\left(\frac{\nu}{|\lambda_{\mathbf{W}}|}\mathbf{W}\boldsymbol{s}_{t-1} + \mathbf{U}\tilde{\mathbf{z}}_{t-\ell}\right).$$

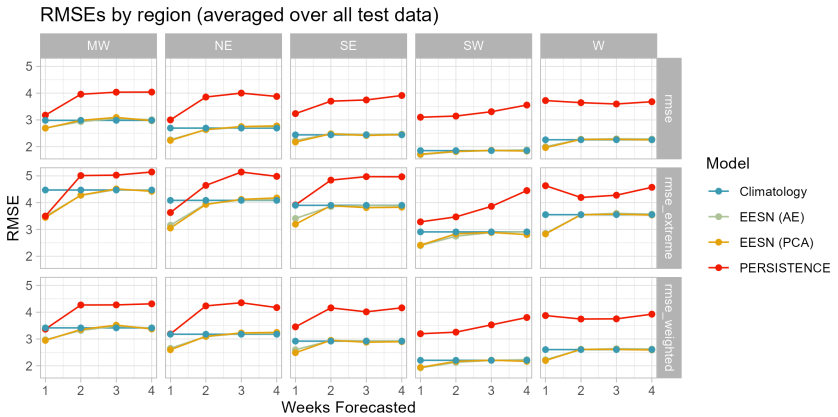
- $\tilde{\boldsymbol{z}}_{t-\ell}$ represents our (possibly lagged) latent inputs.
- $\lambda_{\mathbf{W}}$ is the spectral radius of $\mathbf{W}$, $\nu \in [0, 1]$ controls memory.
- g is a non-linear function (usually $\tanh(\cdot)$).
- $\boldsymbol{W}$ and $\boldsymbol{U}$ are randomly sampled and sparse.

$$\mathbf{W}[h, c_w] = \gamma^w_{h,c_w}\text{Unif}(-a_w, a_w) + (1 - \gamma^w_{h,c_w})\delta_0,$$
$$\mathbf{U}[h, c_u] = \gamma^u_{h,c_u}\text{Unif}(-a_u, a_u) + (1 - \gamma^u_{h,c_u})\delta_0$$

where $\gamma^w_{h,c_w} \sim Bern(\pi_w)$, $\gamma^u_{h,c_u} \sim Bern(\pi_u)$, δ_0 is a Dirac delta at 0, and values of a_w, a_u, π_w, and π_u are pre-specified and set to small values.

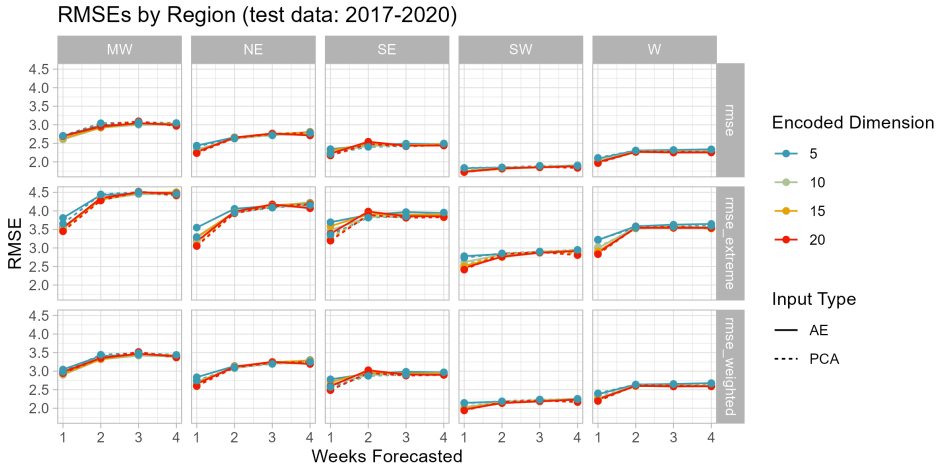
- Only output weights, $\mathbf{B}$, are trained via ridge regression.

Forecast Comparison: PCA vs Autoencoder Features

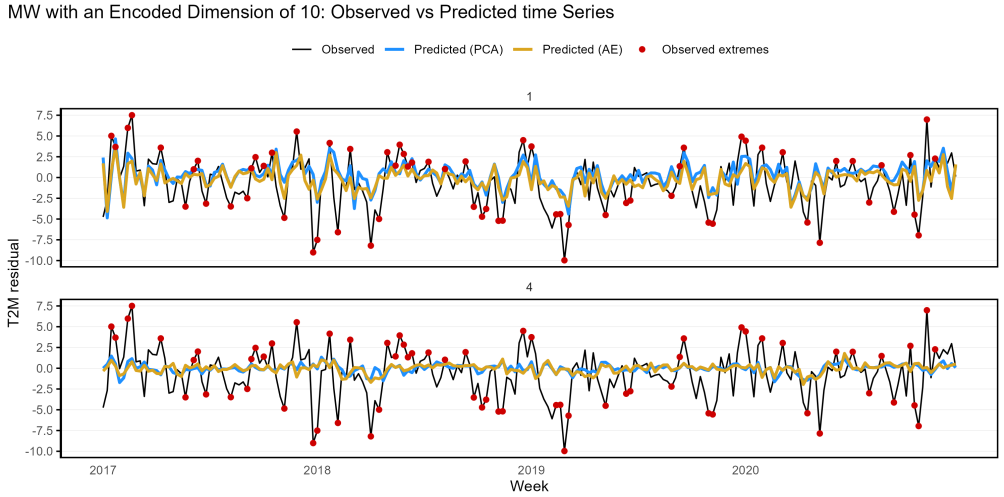


- AE and PCA features give similar forecast RMSE.

Forecast Comparison: PCA vs Autoencoder Features

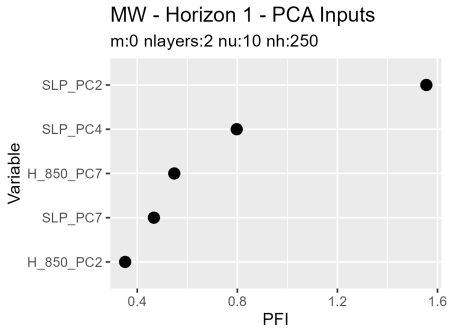


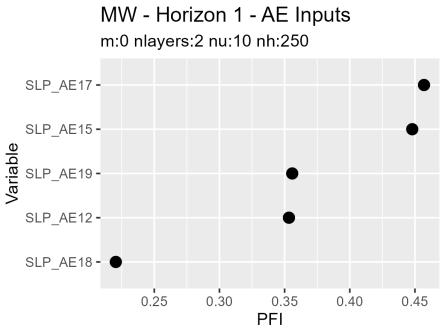
Forecast Comparison: PCA vs Autoencoder Features



Feature Importance and Physical Interpretation

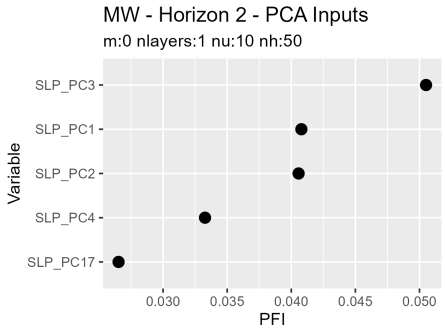
Permutation feature importance (I_G) measures the loss of predictive skill when a feature (or group of features), G, is randomized:

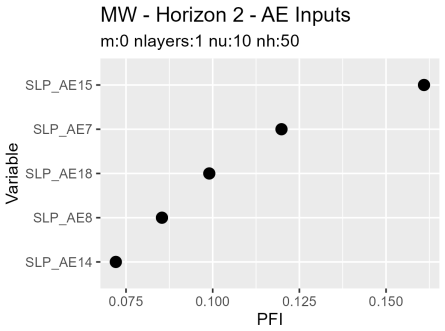




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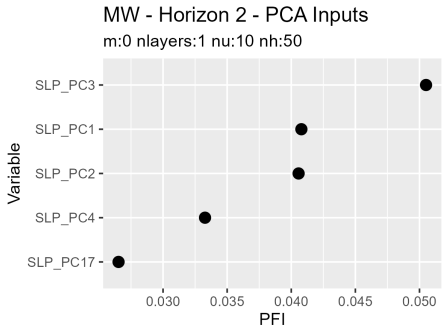
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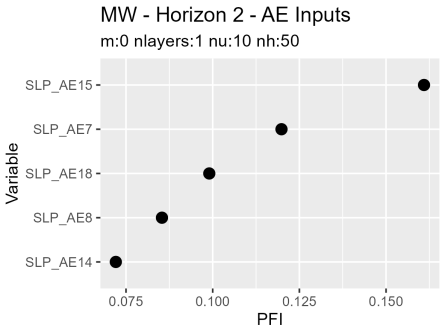




Feature Importance and Physical Interpretation

Permutation feature importance (I_G) measures the loss of predictive skill when a feature (or group of features), G, is randomized:





- Once we know which features are the most important, we can look into their physical interpretation.

Interpreting Latent Features

Each predictor/model input $z^{(m)}$ can be interpreted by its spatial structure.

PCA gives spatial patterns directly:

$$\hat{x}^{(m)}_{t,\mathrm{PCA}} = \bar{x}^{(m)} + \boldsymbol{V}^{(m)} z^{(m)}_t.$$

The columns of $\boldsymbol{V}^{(m)}$ are EOF maps.

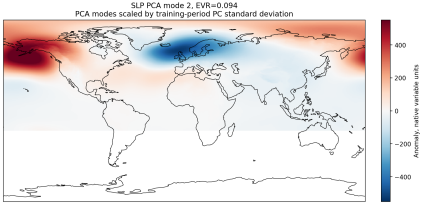
Autoencoders require decoded perturbations:
Define $z^{(m)}_{j,+}$ and $z^{(m)}_{j,-}$ by adding and subtracting $c\sigma^{(m)}_j$ from coordinate j of $\bar{z}^{(m)}$.

- $\sigma^{(m)}_j$: standard deviation of latent coordinate j.
- $\Delta^{(m)}_j \in \mathbb{R}^{P_m}$ can be reshaped into an EOF-like spatial sensitivity map.

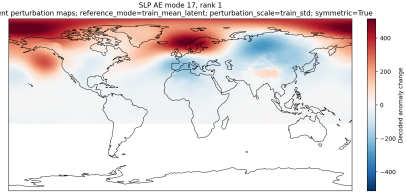
$$\Delta^{(m)}_j = \frac{1}{2c}\left[g_{2,\hat{\phi}_m}\left(z^{(m)}_{j,+}\right) - g_{2,\hat{\phi}_m}\left(z^{(m)}_{j,-}\right)\right]$$

Example: EOFs and Decoded AE Perturbations

Top PCA Input $(h = 1)$

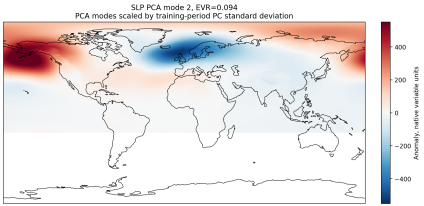


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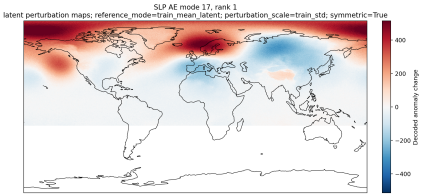


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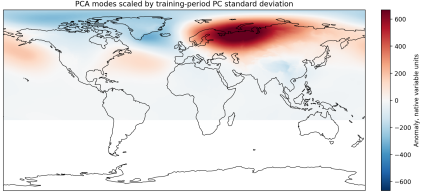
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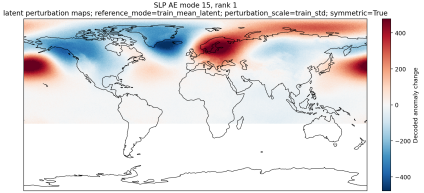
Top AE Input $(h = 1)$



Top PCA Input $(h = 2)$

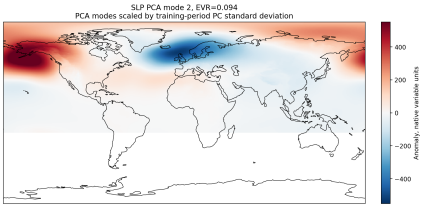


Top AE Input $(h = 2)$

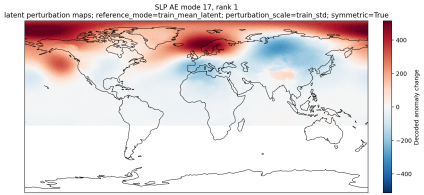


Example: EOFs and Decoded AE Perturbations

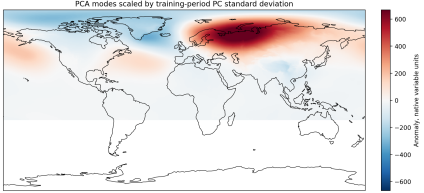
Top PCA Input $(h = 1)$



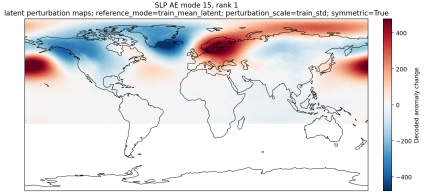
Top AE Input $(h = 1)$



Top PCA Input $(h = 2)$



Top AE Input $(h = 2)$



These maps connect important latent features to physical spatial patterns.

Future Work: Forecast-Aware Autoencoders

Current AE features are optimized for reconstruction:

$$\mathcal{L}_{rec} = \sum_t \|x_t - \hat{x}_t\|^2 .$$

A natural next step is to train latent features with a prediction-aware objective:

$$\mathcal{L} = \sum_t \|x_t - \hat{x}_t\|^2 + \lambda \sum_t \|y_{r,t+h} - \hat{y}_{r,t+h}\|^2 .$$

- Balances reconstruction and forecast relevance.
- Learned z_t can still be used as inputs to RF/EESN.

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- Balances reconstruction and forecast relevance.
- Learned z_t can still be used as inputs to RF/EESN.

An additional goal is to transition to spatial forecasts instead of regional averages

Summary and Takeaways

- Both PCA and autoencoders can be used to create inputs for ML models that improve subseasonal temperature forecasts.
- Autoencoder features currently give comparable forecast RMSE to traditional PCA features.
- Reconstruction-optimized features are not necessarily prediction-optimal:
- Decoded latent perturbations allow for EOF-like interpretability for AE latent features.

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[3] Katherine Goode, Daniel Ries, and Kellie McClernon. "Characterizing climate pathways using feature importance on echo state networks". In: *Statistical Analysis and Data Mining: The ASA Data Science Journal* 17.4 (2024), e11706. DOI: https://doi.org/10.1002/sam.11706. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/sam.11706. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.11706.

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[5] Mio Matsueda and Tetsuo Nakazawa. “Early warning products for severe weather events derived from operational medium-range ensemble forecasts”. In: *Meteorological Applications* 22.2 (2015), pp. 213–222. DOI: https://doi.org/10.1002/met.1444. eprint: https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/met.1444. URL: https://rmets.onlinelibrary.wiley.com/doi/abs/10.1002/met.1444.

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[7] Patrick L. McDermott and Christopher K. Wikle. “Deep echo state networks with uncertainty quantification for spatio-temporal forecasting”. In: *Environmetrics* 30.3 (2019). e2553 env.2553, e2553. DOI: https://doi.org/10.1002/env.2553. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/env.2553. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/env.2553.

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